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1 Nilradical and Jacobson Radical

Let $\mathcal{R}$ be the Jacobson radical of the ring A , ie $\mathcal{R}$ is the intersection of all maximal ideals of A .

Claim: $x \in \mathcal{R}$ iff for all $y \in A$, we have $1 - xy$ is a unit.

Proof: Suppose $x \in \mathcal{R}$. Suppose FTSOC $1 - xy \notin A^\times$. Pick maximal $\mathcal{M}$ containing $1 - xy$. Since $xy \in \mathcal{M}$, we have $1 \in \mathcal{M}$, absurd. Now suppose $1 - xy \in A^\times$ for all y . Suppose FTSOC $\mathcal{M}$ is maximal such that $x \notin \mathcal{M}$. Since $A/\mathcal{M}$ is a field, pick y such that $1 - xy \in \mathcal{M}$. However, $\mathcal{M} \cap A^\times = \emptyset$.

2 Operations on Ideals

Claim: Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be ideals in A . Then $\mathcal{A}(\mathcal{B} + \mathcal{C}) = \mathcal{A}\mathcal{B} + \mathcal{A}\mathcal{C}$.

Proof: $\mathcal{A}(\mathcal{B} + \mathcal{C})$ is generated by elements $a(b + c) = ab + ac \in \mathcal{A}\mathcal{B} + \mathcal{A}\mathcal{C}$, so $\mathcal{A}(\mathcal{B} + \mathcal{C}) \leq \mathcal{A}\mathcal{B} + \mathcal{A}\mathcal{C}$. $\mathcal{A}\mathcal{B} + \mathcal{A}\mathcal{C}$ is generated by elements $a_1b + a_2c \in \mathcal{A}(\mathcal{B} + \mathcal{C})$, so equality holds.

Claim: $\mathcal{A} \cap (\mathcal{B} + \mathcal{C}) = \mathcal{A} \cap \mathcal{B} + \mathcal{A} \cap \mathcal{C}$ over $\mathbb{Z}$ but not in general.

Proof: Take $A = (x + y), B = (x), C = (y)$. The LHS is $(x + y)$. The RHS is 0. Over $\mathbb{Z}$:

$$\begin{aligned} m\mathbb{Z} \cap (n\mathbb{Z} + k\mathbb{Z}) &= m\mathbb{Z} \cap \gcd(n, k)\mathbb{Z} = \text{lcm}(m, \gcd(n, k))\mathbb{Z} \\ &= \gcd(\text{lcm}(m, n), \text{lcm}(m, k))\mathbb{Z} \end{aligned}$$

(assuming degenerate cases such as $n = k = 0$ are checked). We need to check $\text{lcm}(m, \gcd(n, k)) = \gcd(\text{lcm}(m, n), \text{lcm}(m, k))$. Indeed, let α, β, γ denote the power of p in m, n, k (if any are zero, take the power to be ∞). We wish for

$$\max(\alpha, \min(\beta, \gamma)) = \min(\max(\alpha, \beta), \max(\alpha, \gamma)).$$

Claim: (modular law). If $\mathcal{A} \geq \mathcal{B}$ or $\mathcal{A} \geq \mathcal{C}$, then $\mathcal{A} \cap (\mathcal{B} + \mathcal{C}) = \mathcal{A} \cap \mathcal{B} + \mathcal{A} \cap \mathcal{C}$.

Proof: WLOG $\mathcal{A} \geq \mathcal{B}$ and let $x \in \mathcal{A} \cap (\mathcal{B} + \mathcal{C})$, so $x \in \mathcal{A}$ and $x = b + c$ for some b, c . Thus, $b \in \mathcal{A} \cap \mathcal{B}$ and $c \in \mathcal{A} \cap \mathcal{C}$, so $x \in \mathcal{A} \cap \mathcal{B} + \mathcal{A} \cap \mathcal{C}$. We always have $\mathcal{A} \cap \mathcal{B} + \mathcal{A} \cap \mathcal{C} \leq \mathcal{A} \cap (\mathcal{B} + \mathcal{C})$.

Claim: $(\mathcal{A} + \mathcal{B})(\mathcal{A} \cap \mathcal{B}) = \mathcal{A}\mathcal{B}$ in $\mathbb{Z}$ and the $=$ becomes $\leq$ in general.

Proof:

$$(m\mathbb{Z} + n\mathbb{Z})(m\mathbb{Z} \cap n\mathbb{Z}) = (\gcd(m, n)\mathbb{Z})(\text{lcm}(m, n)\mathbb{Z}) = mn\mathbb{Z}$$

$$\begin{aligned} (\mathcal{A} + \mathcal{B})(\mathcal{A} \cap \mathcal{B}) &= \mathcal{A}(\mathcal{A} \cap \mathcal{B}) + \mathcal{B}(\mathcal{A} \cap \mathcal{B}) \\ &\leq \mathcal{A}\mathcal{B} + \mathcal{B}\mathcal{A} = \mathcal{A}\mathcal{B} \end{aligned}$$

Claim: (i) If $\mathcal{A}_i, \mathcal{A}_j$ are coprime for $i \neq j$, then $\prod \mathcal{A}_i = \bigcap \mathcal{A}_i$. (ii) ϕ is surjective is equivalent to $\mathcal{A}_i, \mathcal{A}_j$ are coprime for $i \neq j$. (iii) ϕ is injective is equivalent to $\bigcap \mathcal{A}_i = 0$.

Proof: (i) It suffices to show $\prod_{i=1}^k \mathcal{A}_i$ and $\mathcal{A}_{k+1}$ are coprime for $k \in [n-1]$. The result holds for $k=1$. Suppose for k . Then

$$\begin{aligned} (1) &= \left(\prod_{i=1}^k \mathcal{A}_i + \mathcal{A}_{k+2} \right) (\mathcal{A}_{k+1} + \mathcal{A}_{k+2}) = \prod_{i=1}^{k+1} \mathcal{A}_i + \left(\prod_{i=1}^k \mathcal{A}_i + \mathcal{A}_{k+1} \right) \mathcal{A}_{k+2} + \mathcal{A}_{k+2}^2 \\ &= \prod_{i=1}^{k+1} \mathcal{A}_i + \mathcal{A}_{k+2} + \mathcal{A}_{k+2}^2 = \prod_{i=1}^{k+1} \mathcal{A}_i + \mathcal{A}_{k+2} \end{aligned}$$

completing the induction.

(ii) Suppose ϕ is surjective and $i \neq j$. Pick $a \in \mathcal{A}$ such that $\bar{a} = \bar{1}$ in $\mathcal{A}_i$ and $\bar{a} = \bar{0}$ in $\mathcal{A}_j$, so $1 - a \in \mathcal{A}_i$ and $a \in \mathcal{A}_j$, ie $\mathcal{A}_i + \mathcal{A}_j = (1)$. Conversely, if $\mathcal{A}_i + \mathcal{A}_j = (1)$ for all $i \neq j$, then $\mathcal{A}_i, \prod_{j \neq i} \mathcal{A}_j$ are coprime, so pick $a_i \in \mathcal{A}_i, b_i \in \prod_{j \neq i} \mathcal{A}_j = \bigcap_{j \neq i} \mathcal{A}_j$ such that $a_i + b_i = 1$. Then $\bar{b}_i = \bar{1}$ in $\mathcal{A}_i$ and $\bar{b}_i = \bar{0}$ in $\mathcal{A}_j$ for $j \neq i$. This is enough to show surjectivity.

(iii) Trivial because $\ker \phi = \bigcap_{i=1}^n \mathcal{A}_i$.

Claim: (i) Let $\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_n$ be prime ideals and let $\mathcal{A} \leq \bigcup_{i=1}^n \mathcal{P}_i$. Then $\mathcal{A} \leq \mathcal{P}_i$ for some i . (ii) Let $\mathcal{A}_i$ be ideals and let $\mathcal{P}$ be a prime ideal containing $\bigcap_{i=1}^n \mathcal{A}_i$. Then $\mathcal{P} \geq \mathcal{A}_i$ for some i . If $\mathcal{P} = \bigcap_{i=1}^n \mathcal{A}_i$, then $\mathcal{P} = \mathcal{A}_i$ for some i .

Proof: Consider the case for $n=2$. Sps FTSOC $\mathcal{A} \not\leq \mathcal{P}_1, \mathcal{P}_2$, so pick $a_1 \in \mathcal{A} - \mathcal{P}_1, a_2 \in \mathcal{A} - \mathcal{P}_2$. Since $\mathcal{A} \leq \mathcal{P}_1 \cup \mathcal{P}_2$, we have $a_1 \in \mathcal{P}_2 - \mathcal{P}_1$ and $a_2 \in \mathcal{P}_1 - \mathcal{P}_2$. We have $(a_1, a_2) \leq \mathcal{A} \leq \mathcal{P}_1 \cup \mathcal{P}_2$, so $a_1 + a_2 \in \mathcal{P}_1 \cup \mathcal{P}_2$. WLOG $a_1 + a_2 \in \mathcal{P}_1$, so $a_1 \in \mathcal{P}_1$, absurd.

(i). We proceed by induction on n . The result holds for $n=1$. Suppose the result for n . Let $\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_{n+1}$ be prime ideals. If $\mathcal{A} \leq \bigcup_{j \neq i} \mathcal{P}_j$ for some i , then we're done by the IH. Else, pick $a_i \in \mathcal{A} - \bigcup_{j \neq i} \mathcal{P}_j$, so $a_i \in \mathcal{P}_i$. Define $b_i = \prod_{j \neq i} a_j$ and consider the sum $c = \sum_{i \in [n+1]} b_i \in \mathcal{A}$. WLOG $c \in \mathcal{P}_1$. For $i > 1$ note that b_i has the factor $a_1 \in \mathcal{P}_1$, so $b_i \in \mathcal{P}_1$. Hence, $b_1 \in \mathcal{P}_1$. However $b_1 = \prod_{j \neq 1} a_j$ is a product of factors not in $\mathcal{P}_1$, absurd.

(ii). Sps FTSOC $\mathcal{P} \not\supseteq \mathcal{A}_i$ for all $i \in [n]$. Pick $a_i \in \mathcal{A}_i - \mathcal{P}$ and let $b = \prod_{i \in [n]} a_i$. By absorption, $b \in \bigcap_{i \in [n]} \mathcal{A}_i$. However, each factor isn't in $\mathcal{P}$, so $b \notin \mathcal{P}$, absurd.

3 Modules

Claim: (i). $\text{Ann}(M + N) = \text{Ann } M \cap \text{Ann } N$. (ii). $(N : P) = \text{Ann}((N + P)/N)$.

Proof: $\text{Ann}(M + N) \leq \text{Ann } M \cap \text{Ann } N$. Suppose $a \in \text{Ann } M \cap \text{Ann } N$. Then $a(M + N) = aM + aN = 0$, so $a \in \text{Ann}(M + N)$. $(N : P) = \{a \in A : aP \leq N\} = \{a \in A : a(P + N) \leq N\} = \{a \in A : a((P + N)/N) = 0\}$.

Claim (Nakayama's lemma): Let M be a finitely generated module and a be an ideal contained in the Jacobson radical $\mathcal{R}$. Suppose $aM = M$. Then $M = 0$.

Proof: Pick $x \in A$ such that $x \equiv 1 \pmod{a}$ and $xM = 0$. Suppose FTSOC $x \notin A^\times$. Pick maximal $m \triangleleft A$ containing x . Since $a \leq \mathcal{R} \leq m$, we have $1 \in m$, absurd. Thus, $M = 0$.

Corr: Let M be finitely generated and $a \trianglelefteq \mathcal{R}$ and $N \leq M$ such that $M = aM + N$. Then $M = N$.

Proof: $M/N = (aM + N)/N = a(M/N)$. By the previous claim, $M/N = 0$, so $M = N$.

Claim: Let M be finitely generated over the local ring A with the maximal ideal m . Suppose that the images of $\{x_i\}_{i \in [n]}$ form a $k = A/m$ basis of M/mM . Then the x_i generate M .

Proof: Let $N = \langle x_i : i \in [n] \rangle \leq M$ be a submodule. By assumption, we have $M = mM + N$. Note that $\mathcal{R} = m$ since A is local. Thus, $M = N$.

4 Exact Sequences